MonoArt: Progressive Structural Reasoning for Monocular Articulated 3D Reconstruction

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https://lihaitian.com/MonoArt

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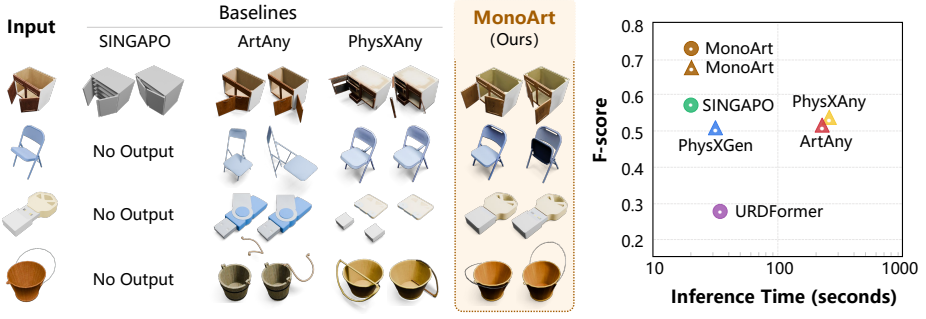

Fig. 1: (Left) Qualitative results of SINGAPO [24], Articulate-Anything (ArtAny) [18], PhysX-Anything (PhysXAny) [4], and MonoArt on diverse objects. **(Right)** F-score vs. inference time on the PartNet-Mobility [51] test set. Circles indicate models evaluated on 7 categories, while triangles denote models supporting all 46 categories.

Abstract. Reconstructing articulated 3D objects from a single image requires jointly inferring object geometry, part structure, and motion parameters from limited visual evidence. A key difficulty lies in the entanglement between motion cues and object structure, which makes direct articulation regression unstable. Existing methods address this challenge through multi-view supervision, retrieval-based assembly, or auxiliary video generation, often sacrificing scalability or efficiency. We present MonoArt, a unified framework grounded in progressive structural reasoning. Rather than predicting articulation directly from image features, MonoArt progressively transforms visual observations into canonical geometry, structured part representations, and motion-aware embeddings within a single architecture. This structured reasoning process enables stable and interpretable articulation inference without external motion templates or multi-stage pipelines. Extensive experiments on PartNet-Mobility demonstrate that MonoArt achieves state-of-the-art performance in both reconstruction accuracy and inference speed. The framework further generalizes to robotic manipulation and articulated scene reconstruction.

Keywords: Monocular Articulated 3D Reconstruction · Progressive Structural Reasoning · Kinematic Estimation

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